

Experiments and Results

Initially

Predicted Bitcoin Price for 22 August 2026: \$68,227.41

Last 30 Days MAPE : 2.07%
Last 30 Days RMSE : \$2527.87
Last 30 Days R^2 : -0.3022

Experiment 01:

13 features

Predicted Bitcoin Price for 22 August 2026: \$66,725.70

Last 30 Days MAPE : 2.07%
Last 30 Days RMSE : \$2527.87
Last 30 Days R^2 : -0.3022

Experiment 02:

12 features : remove the close, 30 days

Predicted Bitcoin Price for 22 August 2026: \$67201.49

MAPE: 2.5156%
RMSE: 2412.74
 R^2 : -0.4688

Experiment 03:

12 features : remove the close + 60 days

Predicted Bitcoin Price for 22 August 2026: \$67,222.09

Last 60 Days MAPE : 7.5882%
Last 60 Days RMSE : \$5253.26
Last 60 Days R^2 : -4.7802

Experiment 04:

13 features : add atr feature

MAPE: 2.5009%
RMSE: 2529.32
 R^2 : -0.6141

Exp 05:

Add roc and remove previous one

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$67712.45

MAPE: 1.9426%

RMSE: 2013.73

R²: -0.0232

Exp 06:

Add close_lag_7

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$68047.71

MAPE: 1.8149%

RMSE: 1826.09

R²: 0.1586

Exp 07:

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$67152.14

MAPE: 3.2800%

RMSE: 2709.21

R²: -0.8519

Exp 08:

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$67094.88

MAPE: 2.2788%

RMSE: 2143.21

R²: -0.1590

Exp 09:

Adding volume roc feature

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$66553.25

MAPE: 3.2863%

RMSE: 2872.14

R²: -1.0814

Exp 10:

Adding EM_20 feature

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$68916.57

MAPE: 2.4375%

RMSE: 2263.32

R^2 : -0.2925

Exp 11:

Add new feature Volatility_7

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$69046.70

MAPE: 2.3884%

RMSE: 2431.56

R^2 : -0.4918

Exp 12:

Add new feature High_Low_Range

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$68441.60

MAPE: 2.5706%

RMSE: 2060.09

R^2 : -0.0708

Exp 13:

Adding new feature SMA_7

MAPE: 1.7997%

RMSE: 1698.75

R^2 : 0.2719

Exp 14:

Adding new feature SMA_14

MAPE: 2.2858%

RMSE: 2226.27

R^2 : -0.2505

Exp 15:

Adding new feature SMA_3

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$67516.92

MAPE: 3.1570%

RMSE: 2814.82

R^2 : -0.9991

Exp 16:

Add new feature Price_SMA7_Ratio

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$68240.83

MAPE: 1.5801%

RMSE: 1544.56

R^2 : 0.3981

Exp 17:

Add new feature Price_SMA7_Deviation -1

STANDALONE LSTM MODEL PREDICTION FOR August 21, 2026: \$67892.50

MAPE: 3.4057%

RMSE: 2875.80

R^2 : -1.0867